

The Universe starts as an unimaginably hot, dense point of energy

In a tiny, tiny fraction of a second, the Universe expands to the size of a city

Over the next 20 minutes, particles called protons and neutrons form, then atomic nuclei

Gravity starts pulling the clouds of hydrogen and helium atoms into clumps

► Formation of the Universe

The time sequence above depicts some key stages in the evolution of the Universe, from the Big Bang, to the formation of atoms, then stars and galaxies, and events through to the present day and into the future. Since the Big Bang, the Universe has cooled and grown larger through the expansion of space itself.

After less than a trillionth of a trillionth of a second, energy starts turning into matter

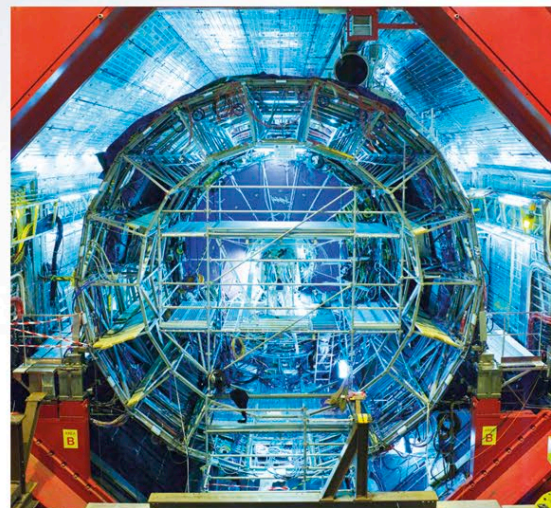
Around 380,000 years later, atoms of hydrogen and helium form

The first stars form after about 550 million years; around the same time, the first galaxies appear

THE BIG BANG

ABOUT 13.8 BILLION YEARS AGO, AN EXCEEDINGLY DRAMATIC EVENT MARKED THE BEGINNING OF BOTH SPACE AND TIME. FROM NOTHING, THE UNIVERSE SUDDENLY APPEARED AS A TINY POINT OF PURE ENERGY.

Within an instant, in what is known as the Big Bang, the Universe expanded trillions of trillions of times, and then continued to get larger, at the same time cooling from its stupendously hot birth. During the first fractions of a second, a vast “soup” of tiny, interacting particles formed out of the intense energy. Some of these joined to make the nuclei (centers) of atoms—the building blocks of everything we can see in the Universe today. Tens of thousands of years later, actual atoms formed and then, after hundreds of millions of years, the very first stars and galaxies.



◄ Studying the Big Bang

Using this complex machine, the Large Hadron Collider, scientists at the European Center for Nuclear Research (CERN) attempt to re-create the conditions that followed the Big Bang. In the collider, beams of high-energy particles are smashed together and the by-products studied.